FOOD ORDERING MANAGEMENT SYSTEM

Product Review and Rating Management System

DBMS PROJECT REPORT

1. INTRODUCTION

The Food Ordering Management System is a database application designed to manage customer details, food products, orders, reviews, and ratings. It uses a relational database to store and retrieve information efficiently. The Product Review and Rating Management System allows customers to share their feedback and rate food products based on their experience. This project demonstrates DBMS concepts such as table creation, SQL queries, constraints, and table relationships.

2. OBJECTIVES

- To maintain customer and food product details.
- To manage food orders efficiently.
- To store customer reviews and ratings.
- To calculate average ratings for food products.
- To retrieve information using SQL queries.
- To maintain data accuracy and consistency using database constraints.
- To apply DBMS concepts in a real-world application.

Create Review and Rating Tables

```
SQL> CREATE TABLE Review(Review_ID VARCHAR2(7), Customer_ID NUMBER(38), Product_ID VARCHAR2(7), Rating NUMBER(1), ReviewText VARCHAR2(50), ReviewDate DATE
PRIMARY KEY(Review_ID), FOREIGN KEY(Customer_ID) REFERENCES Customer(Customer_ID), FOREIGN KEY(Product_ID) REFERENCES Product(Product_ID));

Table created.
```

Store Customer Feedback and Ratings

- **Add reviews for purchased products.**

```
SQL> INSERT INTO Review VALUES ('R101', 101, 'P03', 5, 'Nice food', TO_DATE('2026-09-23','YYYY-MM-DD'));

1 row created.

SQL> INSERT INTO Review VALUES ('R102', 102, 'P01', 4, 'Soft and tasty idli', TO_DATE('2026-05-15','YYYY-MM-DD'));

1 row created.

SQL> INSERT INTO Review VALUES ('R103', 103, 'P02', 5, 'Butter naan is delicious', TO_DATE('2026-03-09','YYYY-MM-DD'));

1 row created.

SQL> INSERT INTO Review VALUES ('R104', 104, 'P05', 4, 'Brownie tastes good', TO_DATE('2026-05-10','YYYY-MM-DD'));

1 row created.

SQL> INSERT INTO Review VALUES ('R105', 105, 'P06', 5, 'Burger is very tasty', TO_DATE('2026-10-10','YYYY-MM-DD'));

1 row created.
```

```
SQL> SELECT * FROM REVIEW;

REVIEW_ CUSTOMER_ID PRODUCT     RATING
------- ----------- ------- ----------
REVIEWTEXT                                                 REVIEWDAT
---------------------------------------------------------- ---------
R101            101 P03              5
The food was very delicious                                23-SEP-26

R102            102 P01              4
Soft and tasty idli                                        15-MAY-26

R103            103 P02              5
Butter naan is delicious                                   09-MAR-26


REVIEW_ CUSTOMER_ID PRODUCT     RATING
------- ----------- ------- ----------
REVIEWTEXT                                                 REVIEWDAT
---------------------------------------------------------- ---------
R105            105 P06              5
Burger is very tasty                                       30-SEP-26
```

- **Store customer ratings from 1 to 5 stars.**

```
SQL> ALTER TABLE Review ADD CONSTRAINT chk_rating CHECK(Rating BETWEEN 1 AND 5);

Table altered.
```

- **Update review comments if required.**

```
SQL> UPDATE Review
  2  SET ReviewText = 'The food was very delicious'
  3  WHERE Review_ID = 'R101';

1 row updated.
```

- **Remove inappropriate or invalid reviews**

```
SQL> DELETE FROM Review WHERE Review_ID='R104';

1 row deleted.
```

Retrieve Product Review Details

- **Display all reviews for a product.**

```
SQL> SELECT * FROM Review WHERE Product_ID='P05';

REVIEW_ CUSTOMER_ID PRODUCT     RATING
------- ----------- ------- ----------
REVIEWTEXT                                                  REVIEWDAT
---------------------------------------------------------- ---------
R104             104 P05                4
Brownie tastes good                                         10-MAY-26
```

- **Display customer name with their reviews.**

```
SQL> SELECT C.Customer_Name,R.ReviewText,R.Rating FROM Customer C JOIN Review R ON C.Customer_ID=R.Customer_ID;

CUSTOMER_NAME
--------------------------------------------------------------------------------
REVIEWTEXT                                          RATING
--------------------------------------------------- ---------
SNEHA
The food was very delicious                              5

VIJAY KUMAR
Soft and tasty idli                                      4

ANITHA
Butter naan is delicious                                 5


CUSTOMER_NAME
--------------------------------------------------------------------------------
REVIEWTEXT                                          RATING
--------------------------------------------------- ---------
KAMESH
Burger is very tasty                                     5
```

- **Find products having maximum reviews**

```
SQL> SELECT Product_ID,COUNT(*) AS Review_Count FROM Review GROUP BY Product_ID ORDER BY Review_Count DESC;

PRODUCT REVIEW_COUNT
------- ------------
P01              1
P06              1
P02              1
P03              1
```

- **Display recent customer feedback.**

```
SQL> SELECT * FROM Review ORDER BY ReviewDate DESC;

REVIEW_ CUSTOMER_ID PRODUCT     RATING
------- ----------- ------- ----------
REVIEWTEXT                                                  REVIEWDAT
----------------------------------------------------------- ---------
R105             105 P06                 5
Burger is very tasty                                        30-SEP-26

R101             101 P03                 5
The food was very delicious                                 23-SEP-26

R102             102 P01                 4
Soft and tasty idli                                         15-MAY-26


REVIEW_ CUSTOMER_ID PRODUCT     RATING
------- ----------- ------- ----------
REVIEWTEXT                                                  REVIEWDAT
----------------------------------------------------------- ---------
R103             103 P02                 5
Butter naan is delicious                                    09-MAR-26
```

- **Retrieve reviews with ratings above 4.**

```
SQL> SELECT * FROM Review WHERE Rating>4;

REVIEW_ CUSTOMER_ID PRODUCT     RATING
------- ----------- ------- ----------
REVIEWTEXT                                                  REVIEWDAT
----------------------------------------------------------- ---------
R101            101 P03                5
The food was very delicious                                 23-SEP-26

R103            103 P02                5
Butter naan is delicious                                    09-MAR-26

R105            105 P06                5
Burger is very tasty                                        30-SEP-26
```

Calculate Average Product Ratings

- **Calculate average rating for each product.**

```
SQL> SELECT Product_ID,AVG(Rating) AS Average_Rating FROM Review GROUP BY Product_ID;

PRODUCT AVERAGE_RATING
------- --------------
P01                4
P03                5
P02                5
P06                5
```

- **Count the number of reviews per product.**

```
SQL> SELECT Product_ID,COUNT(*) AS Number_of_Reviews FROM Review GROUP BY Product_ID;

PRODUCT NUMBER_OF_REVIEWS
------- -----------------
P01                     1
P03                     1
P02                     1
P06                     1
```

- **Find highest-rated products.**

```
SQL> SELECT Product_ID,COUNT(*) AS Number_of_Reviews FROM Review GROUP BY Product_ID;

PRODUCT NUMBER_OF_REVIEWS
------- -----------------
P01                     1
P03                     1
P02                     1
P06                     1
```

- **Identify products with average rating above 4.**

```
SQL> SELECT Product_ID,AVG(Rating) AS Average_Rating FROM Review GROUP BY Product_ID ORDER BY Average_Rating DESC;

PRODUCT AVERAGE_RATING
------- --------------
P03                5
P06                5
P02                5
P01                4
```